Math 556: Graph Theory

Course Syllabus

Virginia Commonwealth University
Fall 2026
Section 001
In person (Face-to-Face Instruction)
T Th 2:00 pm - 3:15 pm
Harris Hall 2101

Instructor

Brent Cody
Office: Harris Hall 4110
Email: bmcody@vcu.edu
Personal Website: https://brentcody.github.io/

Office hours

Harris Hall 4110
Tuesday, 10:00 am - 10:50 am
Thursday, 3:15 pm - 4:15 pm
If these hours don't work for you send me an email to schedule an appointment.

Course description

MATH 556. Graph Theory. 3 Hours.
Semester course; 3 lecture hours. 3 credits. Prerequisites: MATH 211 or MATH 300; MATH 310; and MATH 356, each with a minimum grade of C; or permission of instructor. Introduction to graph classes, graph invariants, graph algorithms, graph theoretic proof techniques and applications.

Course Materials

Required textbook: Introduction to Graph Theory, Second Edition by Douglas West. Published by Prentice Hall, 2001. xx+588 pages, 1296 exercises, 447 figures, ISBN 0-13-014400-2.

- Used copies are available for less than $100.

Course Website

https://learningsystems.vcu.edu/canvas/

Grades

The usual 10-point grading scale will be used to determine course grades (90–100 is an A, 80–89 is a B, etc.). The assignments for the course are broken up into five categories as follows.

Homework	—
Quizzes	20%
Midterm Exams	40% (2 at 20% each)
Presentations	10%
Final Exam	30%
Total	100%

(1) **Homework**: Homework problems will be posted on Canvas by Section. Homework will not be collected or graded, but you should maintain an organized collection of carefully written solutions to homework problems for your own benefit.

(2) **Quizzes (usually on Tuesdays)**: Timed, solo, closed notes, closed book quizzes will be administered in class USUALLY on Tuesdays. Quizzes will typically cover the material from the previous homework assignment.

The first quiz (on Section 1.1) is in class on Tuesday, August 25.

(3) **Midterm Exams (in class)**: There will be two midterm exams in class before the final exam period, each of which I will announce at least one week in advance. Midterm exams must be taken in class during class time (2:00 pm - 3:15 pm) on the specified day, unless you have an accommodation letter.

(4) **Presentations**: During the last week of class each student will give a brief 5-10 minute presentation on a topic/exercise approved by the instructor.

(5) **Final Exam (in class)**: The final exam will be in our usual classroom on

Thursday, December 10, from 12:30 pm to 3:20 pm.

All students must take the final exam.

Grading and makeup policies

- I generally do not distinguish between excused and unexcused absences.
- Makeup quizzes and midterm exams will not be given, even if you have a legitimate excuse for the absence.
- Your lowest 2 quiz grades will be dropped.
- Your lowest midterm exam score will be replaced with your score on the final exam if it improves your course average.
- If you miss one midterm exam, then that counts as your lowest test score, and your final exam score will be used in its place.
- If you miss the final exam for a legitimate reason (illness with doctor's note, etc.), you may receive a grade of incomplete (I).

Expectations and advice

- You belong in this class. I believe that everyone has the ability to learn and master challenging mathematics.
- The only way to truly learn mathematics is to be actively engaged. I therefore expect you to attend all classes, complete all assignments to the best of your ability (including homework!) and ask questions either in class or during office hours.
- It is important that you stay up-to-date with the material being covered in the course; please make use of office hours or class time if anything covered in class is not clear. If the scheduled office hours do not work for you, send me an email to schedule a different time.

In addition to attending the 2 hours and 30 minutes of class each week, I expect that you will spend **at least 8 hours per week outside of class working on this course.**

Help with course material

- Talk to each other as much as possible about course material. Help each other build and understand how to write proofs.
- I enjoy talking to students, so please:
  - ask questions in class,
  - come talk to me in my office and
  - send me email.
- **Resist the temptation to look for solutions to problems.** The best way to learn mathematics is by working through problems yourself. Reading a proof that someone else has written—even if you understand it—does not provide the same experience as developing the proof on your own. Give yourself time to think, experiment, make mistakes, and try different approaches before looking for help. **Struggling with a problem is an important part of learning mathematics.**

Drop Policy

Students who do not attend a single class meeting in the first two weeks of class may be dropped from the class.

Important Dates

Monday, August 24	Last day of add/drop period
Friday, October 16	Midterm grades are due
Friday, October 30	Last day to **withdraw** with a mark of "W".

For other important dates visit: https://academiccalendars.vcu.edu/.

Student learning outcomes

Students in this course are expected to learn how to:

- construct clear, rigorous mathematical proofs and communicate mathematical ideas effectively and
- develop strategies for solving unfamiliar mathematical problems, including the ability to experiment, make conjectures, and use creative approaches.

Student accommodations

VCU is committed to ensuring that all students maintain equal access to all aspects of the university, including educational experiences through the provision of reasonable accommodations and academic adjustments. In addition to being a requirement under Section 504 of the Rehabilitation Act of 1973 and the Americans with Disabilities Act, this speaks directly to VCU's mission of inclusion, equity, and access. To receive accommodations or other disability-related supports, students must register with the Office of Student Accessibility and Educational Opportunity on the Monroe Park Campus (828-2253) or the Division for Academic Success on the MCV campus (828-9782). Students and faculty can visit the Office of Student Accessibility and Educational Opportunity website and/or the Division for Academic Success website for additional information. Once students have completed the registration process, they will be provided with a letter of accommodation. They should provide a copy to their instructor(s) and attempt to schedule a meeting to discuss the implementation of accommodations as early in the semester as possible.

University links and policies

- VCU Syllabus Policy Statements on the Provost's Website
  https://faculty.provost.vcu.edu/faculty-resources/syllabus-statements/
- Use VCU Libraries to find and access library resources, spaces, technology and services that support and enhance all learning opportunities at the university. (https://www.library.vcu.edu/)
- Students should visit http://go.vcu.edu/syllabus and review all syllabus statement information. The full university syllabus statement includes information on safety, registration, the VCU Honor Code, student conduct, withdrawal and more.

Attendance and consequences of poor attendance

The instructional programs at VCU are based upon a series of class meetings involving lectures, discussions, field experiences, special readings and reporting assignments. Therefore it is important for each student to be in attendance on a regular basis. A student who misses a class session is responsible for completing all material covered or assignments made during the absence.

- Students having attendance problems should contact their instructor to explain the reasons for nonattendance and to discuss the feasibility of continuing in the course. If the student has fallen so far behind that the successful completion of the course is impossible, the student should withdraw from the course before the end of the first 10 weeks of classes.
- If the student continues to miss class and does not officially withdraw from the course, the instructor may withdraw the student for nonattendance with a mark of "W" before the end of the first 10 weeks of classes or may assign an academic grade at the end. Withdrawals are not permitted after the end of the first 10 weeks of classes. For classes that do not conform to the semester calendar, the final withdrawal date occurs when half of the course has been completed.

Managing stress

Students may experience situations or challenges that can interfere with learning and interpersonal functioning including stress, anxiety, depression, alcohol and/or other drug use, concern for a friend or family member, loss, sleep difficulties, feeling hopeless or relationship problems. There are numerous campus resources available to students including University Counseling Services (804-828-6200 MPC Campus, 804-828-3964 MCV Campus) which provides brief therapy treatment, University Student Health Services (MPC 804 828-8828, MCV Campus 804 828-9220) and the Department of Recreation & Well-Being (RecWell) (804-828-9355). 24 hour emergency mental health support is available by calling (804) 828-6200 or utilizing the National Suicide Prevention Lifeline (dial 988).